

ILOPANGO, El Salvador

WMO: 786630

Lat: 13.699N

Lon: 89.120W

Elev: 616

StdP: 94.14

Time Zone: -6.00 (W06)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	16.8	17.4	11.0	8.8	21.9	12.4	9.6	21.9	9.2	25.7	8.4	25.8	3.6	0	0.261

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	10.3	33.1	23.1	32.2	23.3	31.9	23.4	26.2	30.2	25.7	29.8	25.4	29.4	3.7	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.1	21.8	29.2	24.5	21.0	28.7	24.1	20.5	28.3	85.8	30.1	83.3	30.3	81.7	29.7	29.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.3	7.2	6.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	25.2	24.5	25.2	26.0	26.3	25.8	25.3	25.5	25.3	25.0	24.7	24.5	24.3
	DBStd	1.17	1.13	0.99	1.03	1.06	0.93	0.95	0.94	0.78	0.90	1.07	1.08	1.16
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	5542	449	426	495	488	491	458	479	475	449	454	435	443
	CDD18.3	2500	191	192	237	238	232	208	221	217	199	196	185	184
	CDH23.3	17233	1441	1507	1952	1836	1557	1291	1565	1438	1112	1054	1193	1287
CDH26.7	6117	571	635	859	718	500	387	557	479	309	289	376	438	
Wind	WSAvg	2.7	3.5	3.1	3.2	2.9	2.3	2.1	2.2	2.2	2.1	2.3	3.2	3.3
Precipitation	PrecAvg	1733	5	4	14	45	153	294	327	322	333	218	34	11
	PrecMax	1992	47	58	70	128	280	443	420	440	520	380	126	100
	PrecMin	1484	0	0	0	0	33	157	232	232	138	84	6	0
	PrecStd	150	11	13	18	40	56	86	74	60	104	96	34	22
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.9	33.7	34.2	34.0	32.5	31.9	32.2	31.9	31.2	31.2	31.5	32.0
		MCWB	23.9	23.1	22.6	23.3	24.2	25.3	24.9	24.8	25.4	25.3	25.1	23.9
	2%	DB	32.0	32.5	33.1	32.7	31.2	30.9	31.4	31.1	30.4	30.2	30.7	31.0
		MCWB	23.0	23.0	22.2	23.4	24.6	24.8	24.4	24.7	25.1	24.7	24.2	23.0
	5%	DB	31.1	31.8	32.2	31.9	30.5	30.0	30.9	30.2	29.8	29.2	29.9	30.1
		MCWB	22.6	22.7	22.2	23.3	24.6	24.7	24.3	24.7	24.9	24.2	23.7	22.5
	10%	DB	30.1	30.8	31.1	30.8	29.7	29.1	29.9	29.7	28.8	28.6	29.0	29.2
		MCWB	22.2	22.2	22.2	23.2	24.5	24.5	24.4	24.6	24.5	23.9	23.2	22.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.7	25.7	25.3	26.1	26.2	26.2	26.3	26.4	26.6	26.4	26.0	25.4
		MCDB	31.1	31.0	30.9	30.6	30.4	30.1	30.3	30.2	30.4	30.1	30.6	30.2
	2%	WB	24.0	24.7	24.4	25.3	25.5	25.6	25.7	25.7	25.7	25.5	25.1	24.3
		MCDB	30.5	30.3	30.6	30.3	29.8	29.5	29.8	29.6	29.4	29.0	29.2	29.4
	5%	WB	23.4	23.9	23.8	24.7	25.1	25.1	25.2	25.2	25.1	24.9	24.4	23.5
		MCDB	29.8	29.9	30.1	29.7	29.2	28.9	29.3	29.1	28.5	28.2	28.4	28.7
	10%	WB	22.8	23.2	23.2	24.0	24.6	24.7	24.7	24.8	24.6	24.3	23.7	22.7
		MCDB	28.9	29.1	29.0	28.7	28.5	28.2	28.6	28.5	27.7	27.4	27.7	27.8
Mean Daily Temperature Range	5% DB	MDBR	12.5	12.7	12.5	10.3	8.8	8.6	9.6	9.4	8.5	8.4	10.1	11.3
		MCDBR	13.3	13.5	13.3	11.4	9.9	9.7	10.6	10.3	9.8	9.6	10.9	11.8
		MCWBR	6.3	6.0	5.8	5.0	4.8	5.0	5.3	5.5	5.2	5.2	5.8	5.8
	5% WB	MCDBR	12.8	13.0	12.3	10.4	9.1	9.1	9.9	9.8	9.1	8.9	10.2	11.5
		MCWBR	6.4	6.6	5.9	5.0	4.6	4.9	5.5	5.5	5.1	5.1	5.8	6.1
Clear-Sky Solar Irradiance	taub	0.344	0.356	0.390	0.477	0.502	0.415	0.417	0.411	0.405	0.383	0.362	0.345	
	taud	2.588	2.554	2.435	2.178	2.133	2.428	2.424	2.432	2.435	2.538	2.571	2.605	
	Ebn at Noon	957	956	925	841	806	870	871	886	898	917	931	948	
	Edh at Noon	96	103	119	153	158	116	117	117	117	104	97	93	
All-Sky Solar Radiation	RadAvg	5.58	6.14	6.48	6.20	5.57	5.62	6.27	6.16	5.50	5.17	5.43	5.36	
	RadStd	0.13	0.20	0.33	0.35	0.41	0.32	0.36	0.41	0.40	0.44	0.20	0.14	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page